

11. EPSILON CLOSURE OF ALL STATES IN AN NFA

=====

```
#include <stdio.h>
```

```
int n, e[20][20], vis[20];
```

```
void dfs(int s) {
```

```
    int i;
```

```
    vis[s] = 1;
```

```
    for(i=0;i<n;i++)
```

```
        if(e[s][i] && !vis[i])
```

```
            dfs(i);
```

```
}
```

```
int main() {
```

```
    int i,j;
```

```
    printf("Enter number of states: ");
```

```
    scanf("%d",&n);
```

```
    printf("Enter epsilon transition matrix:\n");
```

```
    for(i=0;i<n;i++)
```

```
        for(j=0;j<n;j++)
```

```
            scanf("%d",&e[i][j]);
```

```
    for(i=0;i<n;i++) {
```

```
        for(j=0;j<n;j++) vis[j]=0;
```

```
        dfs(i);
```

```
        printf("E-Closure(q%d) = { ",i);
```

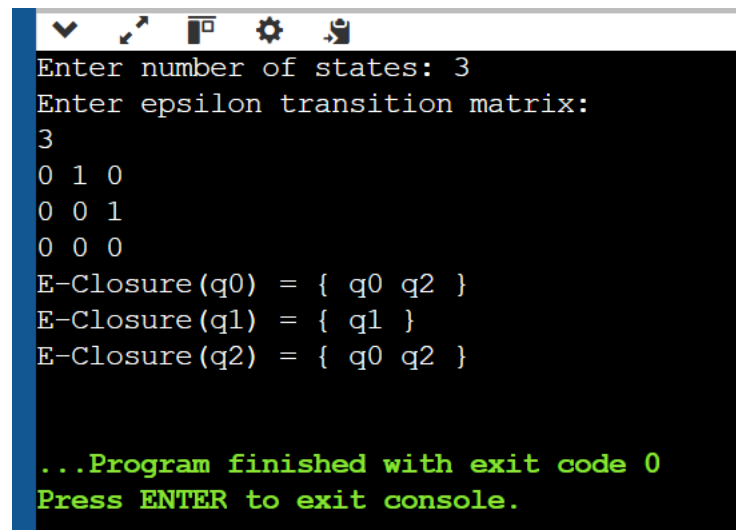
```
        for(j=0;j<n;j++)
```

```
            if(vis[j]) printf("q%d ",j);
```

```
        printf("}\n");
```

```
}
```

```
return 0;  
}
```

A terminal window with a dark background and a light blue title bar. The title bar contains several icons: a checkmark, a cursor, a window icon, a gear, and a magnifying glass. The terminal text is as follows:

```
Enter number of states: 3  
Enter epsilon transition matrix:  
3  
0 1 0  
0 0 1  
0 0 0  
E-Closure(q0) = { q0 q2 }  
E-Closure(q1) = { q1 }  
E-Closure(q2) = { q0 q2 }  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```